RESEARCH PLAN · RANSKNOW DATASET

From keyword extraction to a benchmarked, uncertainty-aware dataset

An experimental plan for a paper on RansKnow: how it was built, what it actually contains, and what a rigorous set of baseline, calibration, conformal, and out-of-distribution experiments looks like on top of it.

1,128	1,034	92	21	92.7%
video records	with transcripts	channels	families detected	ASR-transcribed

CONTENTS

FRAMING

Why this plan, not the usual one

The obvious next step after the Knowledge Agent is "train some classifiers on the features and report accuracy." That paper gets rejected, or worse, gets accepted and is wrong, because the labels the classifiers would be trained on *and* evaluated against are the same regex-matched labels the Knowledge Agent produced — there is no independent ground truth in the loop anywhere. Reporting 90% F1 against your own keyword matcher's opinion of itself is not a result.

So this plan is ordered deliberately: **Phase 0 exists before any modeling**, because two audits of the current labels turned up real, checkable problems that would otherwise quietly become the paper's biggest weakness if a reviewer found them instead of you. Everything after that — the model ladder, calibration, conformal prediction, OOD detection — is standard and well-trodden. The label-quality audit and the small human-annotated evaluation set are the part that makes the rest of it trustworthy, and arguably the part most worth writing about.

SECTION 1

Two findings that reshape the plan

FINDING A — GENERIC-WORD ALIAS COLLISIONS

`Play` is the single most-detected ransomware family in the dataset — 386 of the 494 family-tagged videos (78%) carry it, dwarfing every other family. It is also an English word that appears in ordinary sentences ("let's *play* the video," "just *play* with it") far more often than in ransomware discussions. I manually checked 8 `Play`-tagged videos at random: **zero** were actually about Play ransomware. The same pattern shows up for `Hive` (collides with "registry hive," a routine DFIR term — 2 of 4 spot-checked hits were the registry, not the group) and `Cactus` (collides with CactusCon, the conference — 1 of 4 hits was the conference name). The alias table (`rubrics/Ransomware_Family_Coverage_List.xlsx`) adds the bare family name as a matchable alias for every family, with no requirement for disambiguating context, so any family whose canonical name is also a common word or overlaps a security-adjacent term inherits this failure mode for free.

FINDING B — MOST "TRANSCRIPTS" ARE WHISPER OUTPUT, NOT CAPTIONS

Only 74 of 1,034 transcripts (7.2%) came from official YouTube captions via the API. **959 (92.7%) are ASR transcriptions** — 554 from `whisper_base` (the smallest, least accurate Whisper checkpoint) and 405 from a local Whisper fallback. Aggregate detection rates don't show an obvious gap between providers (family-detection rate is actually similar across all three — 44–49%), so this isn't clearly suppressing recall in the aggregate. But `whisper_base` is known to mis-transcribe rare proper nouns and jargon — exactly the vocabulary the Knowledge Agent's regexes are built to catch (`LockBit`, `Mimikatz`, `rclone`). This needs a targeted spot-check rather than an aggregate-rate argument, because the errors that matter here are the kind that wouldn't move an aggregate rate much while still being individually consequential.

FINDING C — `TRANSCRIPT_PATH` IS NOT PORTABLE

The feature CSV stores absolute filesystem paths from whatever machine happened to run `knowledge_agent.py` at the time (e.g. `/tmp/claude-1004/.../RansKnow_repo2/transcripts/C01.../V0001.txt` next to plain `/home/henrykabuye/RansKnow/transcripts/...` rows in the same column). This is a five-minute fix — regenerate the column as a path relative to the repo root — but it should happen before anyone else tries to reproduce a result from the CSV alone.

SECTION 2

Dataset snapshot

Numbers below are computed directly from `outputs/Knowledge_Agent_Features_1034.csv`, not estimated.

Coverage by year

YEAR	VIDEOS	YEAR	VIDEOS
2016	3	2022	46
2017	1	2023	88
2018	1	2024	180
2019	12	2025	299
2020	29	2026	338
2021	37	**Total**	**1,034**

79% of the dataset (817/1,034) is from 2024 onward — this is a recency-skewed corpus, which is exactly what makes a train-on-old / test-on-new temporal split meaningful rather than an afterthought (see Phase 4).

Family, tactic, platform, tool base rates

SIGNAL	DISTRIBUTION
Family_Count	0 families: 540 (52.2%) · 1: 364 · 2: 84 · 3: 30 · 4: 10 · 5: 2 · 6: 4
Top families	Play 386 · Qilin 93 · Conti 33 · Royal 28 · LockBit 27 · Akira 24 · Hive 23 · Cl0p 11 · Maze 11 · INC 10 (21 unique total)
Dominant_Tactic	Initial_Access 474 · Impact 179 · *none* 122 · Execution 97 · C2 52 · Discovery 32 · others < 30 each
Platform_Signal	*none* 487 · Windows 416 · Linux 81 · ESXi 50
Tool_Total_Mentions	Any tool mentioned: 105/1,034 (10.2%) · median = 0 · max = 54
DurationSeconds	mean 1,914s (~32 min) · median 1,770s · range 300s–17,618s
Transcript_Provider	whisper_base 554 · local_asr_whisper 405 · youtube_api 74 · ytdlp_captions 1

The shape to plan around: family and tool labels are sparse and long-tailed (single-digit counts for most of the 21 families), tactic/platform labels are dominated by two or three majority classes with a large "none detected" bucket, and everything sits on top of a corpus that is unevenly split across 92 channels (up to 20 videos each) and skewed hard toward 2024–2026.

SECTION 3

Task taxonomy

Five predictive tasks fall directly out of the existing schema. All five currently use Knowledge Agent output as their *only* label source — Phase 0 is what earns the right to call any of them ground truth.

TASK	TYPE	CLASSES	LABEL SOURCE TODAY
Ransomware family	multi-label	21	alias regex match

TASK	TYPE	CLASSES	LABEL SOURCE TODAY
Dominant MITRE tactic	multi-class (+none)	11	max of 10 tactic-pattern counts
Platform signal	multi-class (+none)	4	max of 3 platform-pattern counts
Tool mentions	multi-label	8	per-tool regex match
Ransomware-relevant screen	binary	2	Family_Count > 0 *or* Tactic_Impact > 0 (proposed)

The fifth task isn't in the current schema — it's worth adding because a fast, cheap "is this video even about ransomware" screen is the actual bottleneck if you ever want to point the pipeline at a new batch of channels without hand-reviewing every keyword hit, and it's the easiest task to get a clean gold label for.

`Phase 0 — before any modeling`

Data & label-quality audit

0.1 — Fix the mechanical issues

- Regenerate `Transcript_Path` as repo-relative in the feature CSV (Finding C).
- Add a `Channel_Type` column (Vendor / Conference / DFIR-Independent / Government / Consulting) derived from the channel registries — needed for the GroupKFold-by-type split in Phase 1 and the channel-shift OOD setup in Phase 4.

0.2 — Systematically re-audit the alias list

Don't just fix `Play`. Run every alias in `Ransomware_Family_Coverage_List.xlsx` against a common-English-word check (dictionary lookup, or unigram frequency in a general corpus) and flag any alias that is itself a dictionary word with no ransomware-specific qualifier. For each flagged family, require a co-occurring disambiguator within a short window (e.g. `ransomware`, `group`, `gang`, `encrypt` within ±15 tokens) before counting a hit. Re-run extraction and report the before/after family-count table

— the delta on `Play`, `Hive`, and `Cactus` specifically is very likely a headline number in the paper.

> **EXPERIMENT 0.2**
>
> | | |
> |---|---|
> | Method | Dictionary-collision scan over 41 aliases + windowed-context re-extraction |
> | Output | Precision estimate per flagged family, before/after count table |
> | Cost | ~1 day, no annotation needed |

0.3 — Build a gold evaluation subset

Stratify a sample of ~180 videos across three axes — channel type (Vendor / Conference / Independent / Gov), year bucket (≤2022 / 2023–24 / 2025–26), and current `Family_Count` (0 / 1 / 2+) — then have two annotators independently label each video for: does it discuss a ransomware incident (Y/N), which family if any (open text, then reconciled to the alias list), dominant tactic, and platform. Adjudicate disagreements; report Cohen's κ per field. **This subset is held out from all training in every later phase** — it exists only to answer "how good are these labels, and how good are the models against real labels, not against the labels the models were trained to reproduce."

> **EXPERIMENT 0.3**
>
> | | |
> |---|---|
> | Sample | ~180 videos, stratified 4×3×3 |
> | Annotators | 2 independent + adjudication |
> | Report | Cohen's κ, confusion vs. Knowledge Agent labels |
> | Use | Held-out eval only, never training — reused in Phases 1–4 |

0.4 — ASR jargon spot-check

From the gold subset, pull the whisper-transcribed videos and manually check a fixed term list (`LockBit`, `Mimikatz`, `PsExec`, `Cobalt Strike`, `rclone`, `BloodHound`, family names) against the source audio for transcription accuracy. This turns Finding B from a plausible concern into a measured word-error-rate-on-jargon number.

Phase 1

Baseline model ladder

Three feature representations, five model families, evaluated identically across all five tasks. The rule-based Knowledge Agent itself is baseline zero — it's a strong lexical baseline, not a strawman, and should be reported as one.

REPRESENTATION	FEEDS
Structured	tactic/tool/platform counts, duration, year, channel type — the existing Knowledge Agent columns, minus the label being predicted
TF-IDF	bag-of-words over full transcript text, 1–2 grams
Embedding	pooled sentence embeddings over transcript chunks (any small open embedding model run locally)

MODEL	INPUT	ROLE
Knowledge Agent (rule-based)	raw text	baseline 0
Logistic Regression	structured or TF-IDF	linear baseline
Random Forest	structured or embedding	the model you asked for — also the easiest source of a free uncertainty signal (Phase 2)
Gradient-boosted trees	structured or embedding	usual tabular ceiling
Fine-tuned compact transformer	raw text	text-native ceiling

Three validation protocols, run in parallel

- **Stratified random k-fold** — the standard number, and the one most likely to be optimistic.
- **GroupKFold by `Channel_ID`** — no channel appears in both train and test. With up to 20 videos per channel and only 92 channels, a model can otherwise learn "this is CrowdStrike's narration style" instead of "this transcript discusses lateral movement." The gap between this score and the random-fold score *is* a result worth reporting on its own.

- **Temporal split** — train on pre-2024 (≈217 videos), test on 2024–2026. This is the honest version of "does this generalize," given how young most of the corpus is.

Phase 2

Uncertainty quantification

The point of this phase is comparing *calibration against the Phase 0 gold set*, not against the weak labels — a model that's perfectly calibrated against its own training signal has just learned to be confidently correlated with a regex, which isn't the claim you want in a paper.

EXPERIMENTS

Calibration	Reliability diagrams + Expected Calibration Error, per task, per model, scored against gold labels
Random Forest	tree-vote entropy / class-probability spread as a free uncertainty proxy
Transformer	MC-Dropout at inference and/or a 3–5 seed deep ensemble for predictive entropy
Headline metric	"confidently wrong" rate — fraction of gold-set errors made with >90% predicted confidence, compared across models

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Phase 3

Conformal prediction

Split conformal prediction turns any of the Phase 1 classifiers into a set-valued predictor with a distribution-free coverage guarantee — a natural fit here precisely *because* the point labels are weak: instead of trusting a single predicted family, you get "the true family is in this set with 90% probability," which is a more honest thing to claim about noisy labels.

EXPERIMENTS

Single-label tasks	Split conformal on Dominant_Tactic and Platform_Signal at 90% target coverage — report empirical coverage and mean set size
Multi-label tasks	Per-label conformal calibration (or conformal risk control on Hamming/false-negative rate) for Family and Tool tasks
Conditional coverage stress-test	Recompute coverage restricted to: the gold subset only, post-2025 videos only, conference-channel videos only. If coverage collapses on any slice, that slice *is* the interesting result — it's telling you where the exchangeability assumption breaks, which links straight into Phase 4.

Phase 4

Out-of-distribution detection

Four OOD setups, each grounded in a real split already present in the data rather than a synthetic one.

SETUP	ID	OOD
Temporal emergence	train pre-2024	test videos mentioning families that only appear post-2023 (Qilin, DragonForce, Black Basta)
Channel-type shift	Vendor + DFIR channels	Conference + Independent-YouTuber channels
Leave-rare-family-out	families with n≥10	the 11 families with n<10 held out entirely from training
External non-security corpus	RansKnow transcripts	a general tech-YouTube corpus, as a sanity check that OOD scores actually spike on genuinely unrelated content

METHODS COMPARED, PER SETUP

Baseline	max-softmax-probability
Distance-based	Mahalanobis distance on pooled embeddings
Energy-based	energy score

Conformal-native	nonconformity score from Phase 3 as an OOD flag directly — ties the two phases together instead of treating them as separate chapters
Metric	AUROC and FPR@95%TPR for ID-vs-OOD separation

Phase 5

Confound & robustness checks

- **Length confound** — re-run family/tactic detection as mentions-per-1,000-words rather than raw counts. Longer videos have more chances to trip a keyword regardless of actual content; if the "detection rate" story changes under length normalization, raw counts were partly measuring video length.
- **Channel-identity leakage** — strip or mask channel-identifying phrases (channel name, host names, recurring intros) from transcripts and re-score the Phase 1 models. A large accuracy drop means the model was partly reading "which channel is this" rather than "what does this transcript describe."
- **Provider-stratified error analysis** — report every headline metric split by `Transcript_Provider` (ASR vs. official caption), using the Phase 0.4 jargon spot-check to interpret any gap.

REFERENCE

Metrics by task

PHASE	METRIC
Baselines (multi-label: Family, Tool)	macro-F1, per-class precision/recall (report the long tail, not just macro average)
Baselines (multi-class: Tactic, Platform)	macro-F1, confusion matrix

PHASE	METRIC
Baselines (binary screen)	PR-AUC (base rate is unbalanced — 52% zero-family)
Uncertainty	ECE, reliability-diagram slope, confidently-wrong rate
Conformal	empirical coverage vs. target, mean/median set size, conditional coverage by slice
OOD	AUROC, FPR@95%TPR, per setup and per method

SECTION 12

Suggested paper structure

1. **Introduction** — the gap: no open, transcript-level ransomware knowledge corpus with MITRE-aligned annotations; RansKnow's construction pipeline as the contribution.
2. **Related work** — ransomware threat-intel datasets, weak/distant supervision in security NLP, conformal prediction and OOD detection in NLP.
3. **Dataset construction** — the fetch pipeline, keyword-matched channel curation, the Whisper/caption transcript cascade (own Finding B as a methods-section admission, not a limitations-section footnote).
4. **Label-quality audit** — Findings A–C, the alias re-audit, the gold-subset construction and inter-annotator agreement. This section is the paper's methodological backbone.
5. **Task definitions & baselines** — Phase 1, all three CV protocols reported side by side.
6. **Uncertainty & conformal prediction** — Phases 2–3.
7. **Out-of-distribution robustness** — Phase 4.
8. **Limitations & ethics** — YouTube-sampling bias, ASR noise, weak-label ceiling, dual-use considerations for a ransomware-tactics corpus.
9. **Conclusion** — release the gold-eval subset and audit code as the dataset's second artifact, not just the transcripts.

SECTION 13

Sequencing

PHASE	DEPENDS ON	BLOCKS
0 — data & label audit	nothing	everything below
1 — baseline ladder	Phase 0 gold set	2, 3, 4
2 — uncertainty	Phase 1 trained models	3 (shares calibration infra)
3 — conformal prediction	Phase 1 models, Phase 2 calibration	4 (nonconformity score reused)
4 — OOD detection	Phase 1 embeddings, Phase 3 scores	writing
5 — confound checks	Phase 1 models	can run in parallel with 2–4

Phase 0 is the only true bottleneck — nothing downstream is trustworthy without the gold subset, so it's worth front-loading even though it's the least glamorous phase to write about. Phase 5 can run anytime after Phase 1 finishes and doesn't block anything.